Major Project – 3 Report

1. Project Title:

Accuracy Improvement Challenge in College Admission Prediction Model

2. Project Objective:

The objective of this project is to improve the predictive performance of an existing admission prediction model developed during the class case study. The project focuses on enhancing the baseline model using feature engineering and preprocessing techniques, and evaluating whether these improvements lead to better model accuracy.

3. Methodology:

The project was carried out in multiple stages to systematically evaluate and improve the model.

Baseline Model

The baseline model was built using **Logistic Regression** with three input features:

- GRE score
- GPA
- University Rank

The dataset was split into training and testing sets, and the Logistic Regression model was trained to predict whether a student would be admitted.

The baseline model achieved an accuracy of **0.6625**.

First Improvement Attempt – Feature Engineering

A new feature **GRE_GPA** was created by multiplying the GRE score and GPA to capture the interaction between academic performance indicators.

However, after training the model with this engineered feature, the accuracy decreased to **0.6375**. This suggests that the engineered feature did not provide additional predictive value and may have introduced redundancy or noise into the dataset.

Second Improvement Attempt – Feature Scaling

To improve model performance, **feature scaling** was applied using **StandardScaler**. Logistic Regression models often perform better when input features are normalized, as scaling ensures that all variables contribute proportionally during the optimization process.

After applying feature scaling and retraining the Logistic Regression model, the model achieved an accuracy of **0.675**, which was higher than the baseline accuracy.

4. Model Comparison:

The following table summarizes the performance of the different model variations tested during the project.

Model	Accuracy
Baseline Logistic Regression	0.6625
Logistic Regression + Feature Engineering	0.6375
Logistic Regression + Feature Scaling	0.675

The results show that feature engineering did not improve the model performance, while feature scaling resulted in the highest accuracy among the tested models.

5. Accuracy Improvement:

The improvement in accuracy was calculated relative to the baseline model.

Baseline Accuracy
= **0.6625**

Improved Accuracy
= **0.675**

Accuracy Improvement:

$$\frac{(0.675-0.6625)}{0.6625} \times 100$$

Accuracy Improvement ≈ **1.89%**

Although the improvement is modest, it demonstrates that proper preprocessing techniques such as feature scaling can enhance the performance of machine learning models.

6. Key Insights:

- Logistic Regression performed effectively on this dataset due to the relatively simple relationships between features.
- Feature engineering did not improve performance, indicating that the original features already captured the necessary predictive information.
- Feature scaling helped the model learn more balanced feature contributions, leading to a small but positive accuracy improvement.
- Experimentation with preprocessing techniques is important for improving machine learning model performance.

7. Conclusion:

This project demonstrated how the performance of an existing machine learning model can be improved through experimentation with preprocessing and feature engineering techniques. The baseline Logistic Regression model achieved an accuracy of **0.6625**, while the improved model with feature scaling achieved **0.675**, resulting in an improvement of approximately **1.89%**.

The results highlight the importance of evaluating different improvement strategies and understanding how preprocessing methods can influence model performance. Even small improvements can be valuable in real-world predictive systems.